International Cybersecurity and Digital Forensics Academy (ICDFA)

CIP-B101: Basic Computer Skills for Digital Forensics

Laboratory Report

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

Course code	WADF201
Lab title	Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis
Full name	RISA THOMAS KOILA
Student ID	C11/26/FCDF/17156
Date	7 November 2026
Platform used	HP Laptop 15-bs0xx (Windows 11 Pro, host name DESKTOP-OSED05U) and a Kali Linux VirtualBox VM

1. Introduction

This laboratory is a preliminary, non-invasive examination of a computer that could later become part of a digital investigation. The purpose is to profile the system (hardware, operating system, memory and storage), to map where digital evidence can exist in each system layer, and to observe running programs, processes and PIDs. Disk partitions are then inspected with read-only tools, their sizes are calculated by hand from sector information, and the results are verified against what the operating system reports. The report also explains the PC boot process, compares MBR and GPT, and identifies the file systems found. Two systems were examined: the Windows 11 host (GPT disk, UEFI) and a Kali Linux virtual machine (MBR disk), which gives a real contrast between the two partitioning schemes.

2. Tools Used and Evidence-Handling Statement

Item	Detail
Operating systems	Windows 11 Pro, version 10.0.22621 build 22621 (host); Kali Linux in VirtualBox (guest)
Windows tools	System Information (msinfo32), Command Prompt, PowerShell, DiskPart (list/select only)
Windows commands	systeminfo, wmic diskdrive get model,size,serialnumber, wmic logicaldisk get caption,filesystem,size,freespace,volumename, tasklist, Get-Disk, Get-Partition, Get-Volume, diskpart: list disk / select disk 0 / list partition
Linux commands	lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT, sudo fdisk -l, sudo parted -l, blkid, df -h, lsblk -f, df -Th
Calculation	Manual step-by-step working in this report, checked with a calculator / Python

3. Part A - System Identification and Evidence Layer Mapping

3.1 System specification (Evidence 1)

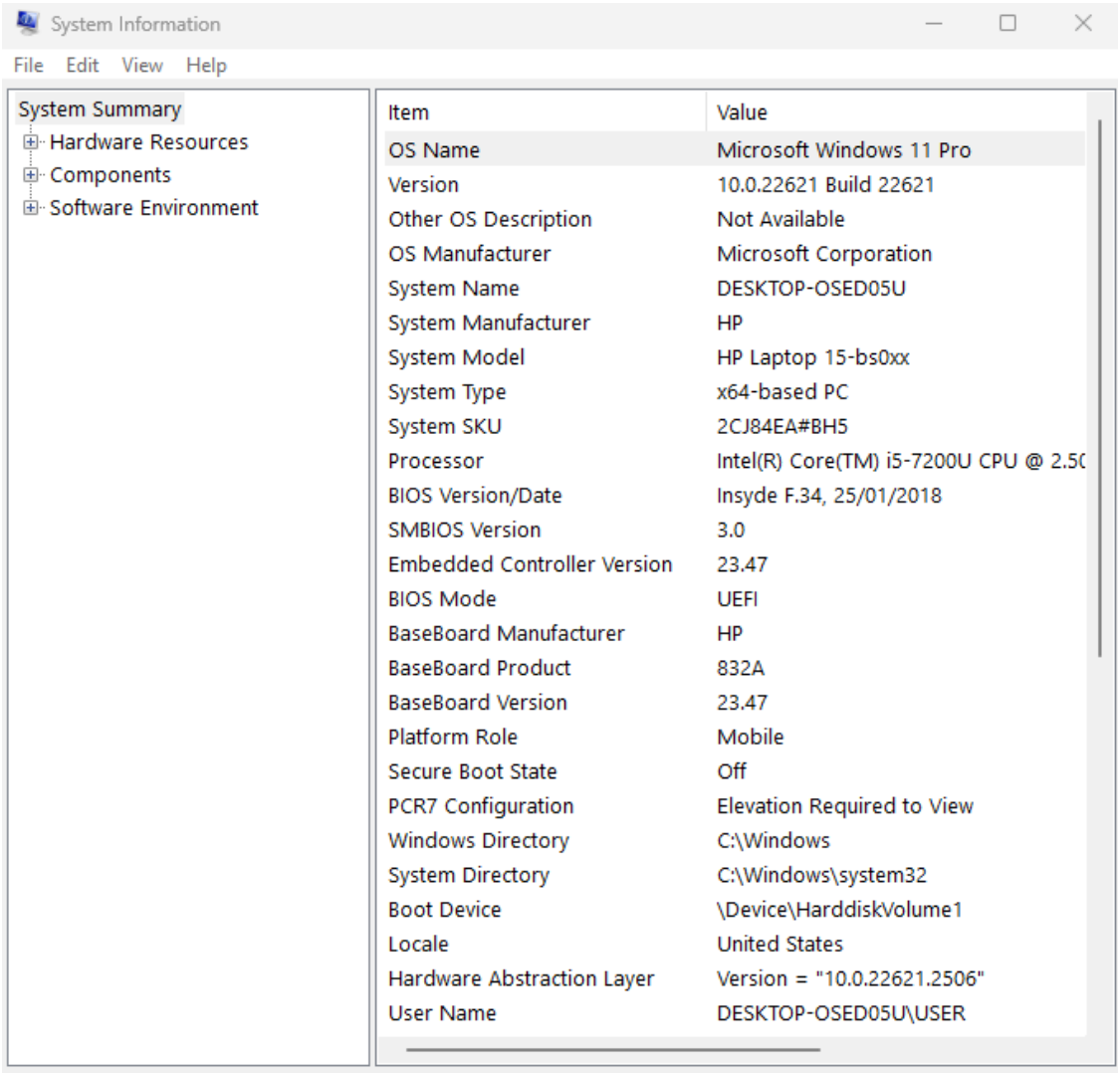


Figure 1 - A1: msinfo32 System Summary (OS, system type, processor, BIOS mode, Secure Boot) [Evidence 1]

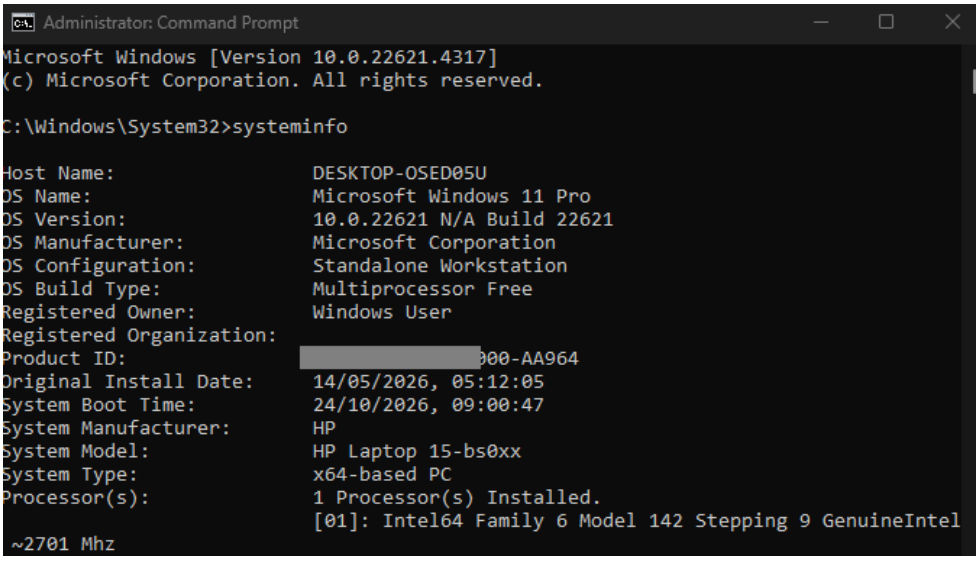


Figure 2 - A1: systeminfo - page 1 (host, OS, model, processor) [Evidence 1]

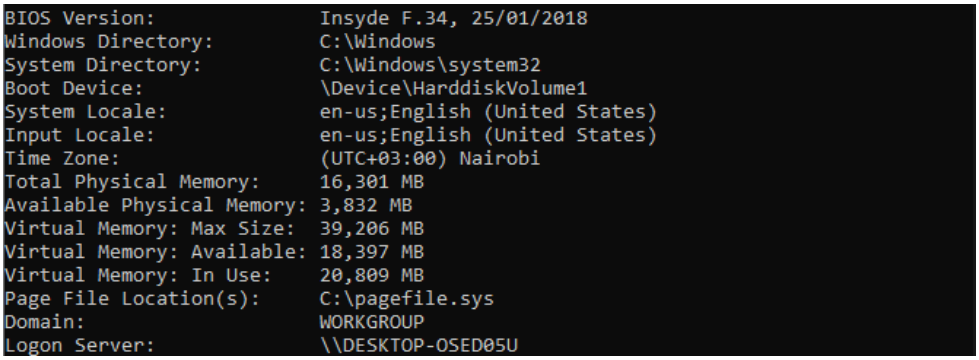


Figure 3 - A1: systeminfo - page 2 (BIOS, boot device, time zone, memory) [Evidence 1]

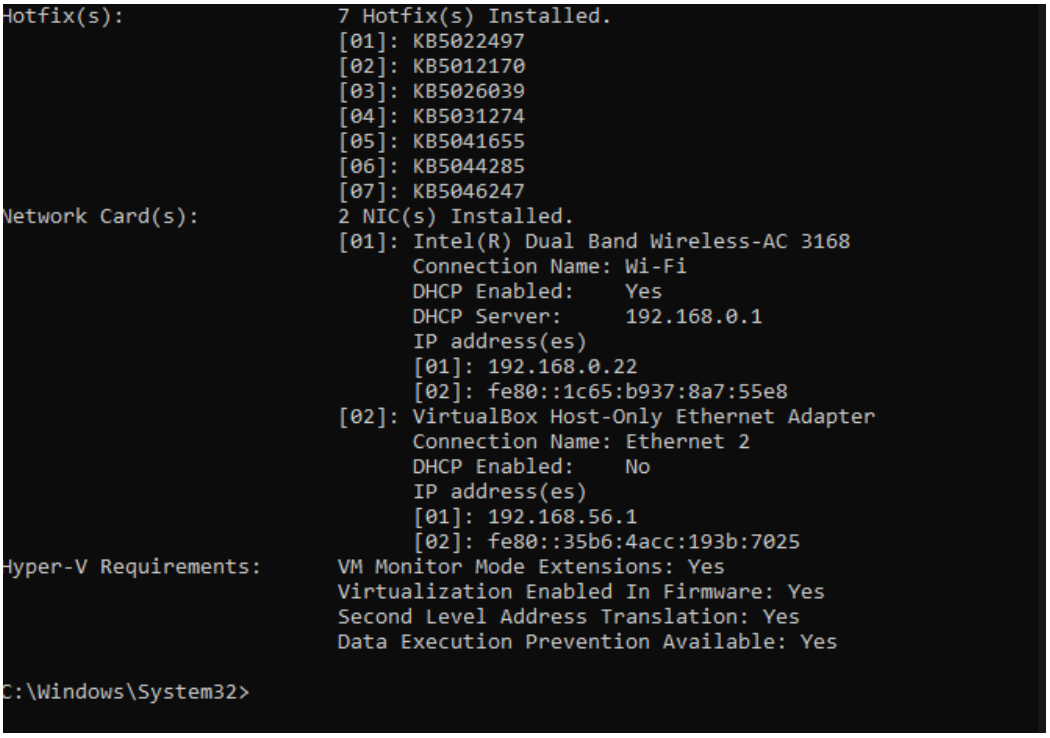


Figure 4 - A1: systeminfo - page 3 (hotfixes, network cards, virtualization) [Evidence 1]

Attribute	Value observed	Source
OS name / version	Microsoft Windows 11 Pro, 10.0.22621 Build 22621	Fig 1, 2
System manufacturer / model	HP – HP Laptop 15-bs0xx (Mobile)	Fig 1, 2
System type	x64-based PC	Fig 1, 2
Processor	Intel Core i5-7200U, ~2.5 GHz base (reported ~2701 MHz), 1 processor	Fig 1, 2
Installed physical memory	16,301 MB (≈ 16 GB); 3,832 MB available at capture time	Fig 3
Virtual memory / page file	Max 39,206 MB; page file C:\pagefile.sys	Fig 3
BIOS version	Insyde F.34, 25/01/2018	Fig 1, 3
BIOS mode	UEFI	Fig 1
Secure Boot state	Off	Fig 1
Boot device	\Device\HarddiskVolume1	Fig 1, 3
Time zone / install date / last boot	UTC+03:00 Nairobi; installed 14/05/2026; booted 24/10/2026 09:00:47	Fig 2, 3
Network	Intel Dual Band Wireless-AC 3168 (Wi-Fi, DHCP); VirtualBox Host-Only Ethernet Adapter	Fig 4
Virtualization	VM Monitor Mode Extensions, Virtualization Enabled in Firmware, SLAT: all Yes	Fig 4

Observations. The machine is a 64-bit Windows 11 laptop with 16 GB of RAM, so a memory capture would be large and should be planned for. It is booted in UEFI mode with Secure Boot off. The VirtualBox host-only adapter and enabled virtualization show that virtual machines run on this host, so the VM and its disk image would be additional evidence sources. The system clock, time zone and last boot time are recorded because they anchor any timeline built from this machine's logs.

3.2 Storage information (Evidence 2 and 3)

```
C:\Windows\System32>wmic diskdrive get model,size,serialnumber
Model                     SerialNumber     Size
Acer SSD RE100 2.5 512GB  [redacted]       512105932800


C:\Windows\System32>
```

Figure 5 - A2: wmic diskdrive get model,size,serialnumber (serial masked) [Evidence 2]

```
C:\Windows\System32>wmic logicaldisk get caption,filesystem,size,freespace,volumena
me
Caption  FileSystem  FreeSpace    Size          VolumeName
C:       NTFS        71768514560  250298232832
D:       NTFS        81997361152  261031456768  Local Disk:
E:


C:\Windows\System32>
```

Figure 6 - A2: wmic logicaldisk get caption,filesystem,size,freespace,volumename [Evidence 3]

Item	Value	Calculation / note
Disk model	Acer SSD RE100 2.5 512GB	Single physical disk, Disk 0
Capacity	512,105,932,800 bytes	÷ 1024^3 = 476.94 GiB (DiskPart shows 476 GB); ÷ 10^9 = 512.11 GB (marketing "512GB")
C:	NTFS, size 250,298,232,832 B (233.11 GiB)	Windows system volume
D:	NTFS, size 261,031,456,768 B (243.10 GiB), label "Local Disk"	Data volume
E:	No file system / size reported	CD-ROM drive with no media (Get-Volume: DriveType CD-ROM)

Observation. The free-space figures for C: differ between the two captures of the same command (Fig 6: 71,768,514,560 B; Fig 19: 71,302,930,432 B). A running system constantly writes to its disk, which is exactly why a live disk is only observed read-only and why the capture time must be recorded.

3.3 Evidence-layer mapping

User activity produces evidence across Applications + Operating System + Disk/Memory. Each layer is shown with what it could yield on this particular machine.

Layer / source	Example evidence	Where found (this system)	Why it matters
User layer	Opened files, search terms, typed commands, logons	Recent items, Jump Lists, browser and PowerShell history under the "USER" profile (Fig 1)	Links a person's actions to system activity
Application layer	Browser cache and cookies, Office metadata, chat logs, VirtualBox VM configuration	%AppData%, %LocalAppData%, VirtualBox folders (host-only adapter, Fig 4)	Shows what software created or handled data
Operating-system layer	Event logs, registry, installed updates, prefetch, accounts	C:\Windows\System32\winevt\Logs, registry hives; 7 hotfixes KB5022497… (Fig 4)	Records logons, execution, configuration changes
File-system layer	MACB timestamps, NTFS $MFT, $UsnJrnl, permissions	C: and D: NTFS volumes (Fig 6, 20)	Builds timelines; shows creation/modification/deletion
Disk / storage layer	Deleted files, slack, unallocated space, hidden partitions	Acer SSD, GPT with 5 partitions incl. EFI, MSR and Recovery (Fig 10, 14)	Recovery of data no longer listed
Memory / process layer	Running malware, PIDs, network connections, keys	16 GB RAM, C:\pagefile.sys (Fig 3), running processes (Fig 7)	Volatile – captured first in live triage
Firmware / hardware layer	UEFI settings, boot order, Secure Boot state	Insyde UEFI F.34, Secure Boot Off (Fig 1)	Explains boot behaviour; may reveal tampering
Network / connected devices	Wi-Fi profiles, DHCP	Intel Wi-Fi 192.168.0.22; host-only 192.168.56.1 (Fig 4)	Extends investigation beyond one machine

	leases, VM virtual NICs		

Why evidence exists in more than one layer. One user action leaves traces at several levels at once. Opening a downloaded document creates a browser history entry (application), a Recent-items shortcut and prefetch record (operating system), timestamps and an MFT entry (file system), the file's data on the disk (storage), and the running process with the document in RAM (memory). Matching the same event across independent layers makes a finding more reliable, preserves a trace if one layer is wiped or altered, and exposes tampering when layers disagree.

4. Part B - Program, Process and Thread Investigation

```
C:\Windows\System32>tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0     23,032 K
Registry                       112 Services                   0     34,508 K
smss.exe                       544 Services                   0        936 K
csrss.exe                      772 Services                   0      3,652 K
wininit.exe                    864 Services                   0      2,164 K
csrss.exe                      872 Console                    1      4,572 K
winlogon.exe                   968 Console                    1      6,516 K
services.exe                   988 Services                   0      8,320 K
lsass.exe                      492 Services                   0     24,488 K
svchost.exe                   1032 Services                   0     46,976 K
fontdrvhost.exe               1060 Console                    1      8,596 K
fontdrvhost.exe               1068 Services                   0      1,244 K
svchost.exe                   1156 Services                   0     20,408 K
svchost.exe                   1216 Services                   0      5,160 K
svchost.exe                   1332 Services                   0      6,664 K
svchost.exe                   1340 Services                   0      2,656 K
svchost.exe                   1348 Services                   0      3,556 K
svchost.exe                   1432 Services                   0      5,640 K
svchost.exe                   1444 Services                   0      4,784 K
svchost.exe                   1512 Services                   0      9,748 K
IntelCpHDCPSvc.exe            1580 Services                   0      2,204 K
svchost.exe                   1588 Services                   0     15,448 K
svchost.exe                   1616 Services                   0      4,796 K
dwm.exe                       1680 Console                    1    153,024 K
svchost.exe                   1712 Services                   0      4,676 K
svchost.exe                   1764 Services                   0      2,448 K
svchost.exe                   1784 Services                   0      4,560 K
IntelCpHeciSvc.exe            2032 Services                   0      2,528 K
svchost.exe                   2040 Services                   0      7,468 K
svchost.exe                   2052 Services                   0      5,928 K
svchost.exe                   2140 Services                   0     11,328 K
svchost.exe                   2192 Services                   0      3,544 K
svchost.exe                   2224 Services                   0     10,196 K
igfxCUIService.exe            2248 Services                   0      3,148 K
svchost.exe                   2256 Services                   0      2,504 K
svchost.exe                   2372 Services                   0      6,520 K
```

Figure 7 - B1: tasklist - running processes with PID and session (first page of output) [Evidence 4]

Process name	PID	Session	Memory	Likely purpose
lsass.exe	492	Services (0)	24,488 K	Local Security Authority: authenticates logons and enforces security policy; a prime target for credential theft, so an unexpected copy or parent is suspicious
winlogon.exe	968	Console (1)	6,516 K	Manages interactive logon/logoff and the secure attention sequence (Ctrl+Alt+Del)
dwm.exe	1680	Console (1)	153,024 K	Desktop Window Manager: composes the graphical desktop; largest memory user on this page of output

Other observations. PID 0 is the System Idle Process and PID 4 is the System process (kernel). Many svchost.exe instances run in session 0 hosting Windows services. Interactive processes (winlogon, dwm, fontdrvhost, one csrss) run in session 1 "Console", while background services run in session 0. PIDs are assigned at run time and change after every reboot, so the capture time matters.

4.2 Program, process and thread

Term	Meaning (own words)	Forensic angle
Program	A passive set of instructions stored as a file on disk (for example lsass.exe in C:\Windows\System32). It does nothing until started.	Examined statically: hash, signature, path, metadata.
Process	A program that is running, with its own PID, memory space, handles and loaded modules (for example PID 492).	Exists only live; shows parent/child relationships, owner, session and network use.
Thread	The smallest unit of execution inside a process; a process can run many threads that share its memory.	Unexpected or injected threads inside a legitimate process are a typical malware sign.

Why process information matters. Process data is volatile and is lost at shutdown, so live triage records it first. A process list can reveal programs with look-alike names (such as "svch0st.exe"), system processes running from the wrong folder or session, wrong parent processes, or processes with unexplained network connections. The PID ties a suspicious process to its file on disk, its memory image and its sockets so each can be cross-checked.

4.3 Why computer forensics is technically difficult

- Many operating systems and versions, each storing artifacts differently (this lab alone involved Windows and Linux).
- Very many applications with their own data formats and databases.
- Varied hardware, firmware and storage technology (here a UEFI/GPT SSD and a virtual MBR disk).
- Connected devices and virtual machines (VirtualBox adapter, Fig 4) that hold related evidence elsewhere.
- Very large data volumes (≈ 512 GB disk, 16 GB RAM) that slow collection and searching.
- Volatile data (processes, RAM) that can be lost or altered just by touching a live system.

5. Part C - Disk Sector and Disk Size Calculation

5.1 Slide calculation (C1)

Formula: size = number of sectors × bytes per sector; GB = bytes ÷ 1024 ÷ 1024 ÷ 1024. Given: 400 sectors per track, 12 heads, 17,000 cylinders, 512 bytes per sector.

Step	Working	Result
Total sectors	400 × 12 × 17,000 = 4,800 × 17,000	81,600,000 sectors
Total bytes	81,600,000 × 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 ÷ 1024	40,800,000 KB
Megabytes	40,800,000 ÷ 1024	39,843.75 MB
Gigabytes	39,843.75 ÷ 1024	38.91 GB

5.2 Method for the real partition table (C2)

1. Start sector = Offset (bytes) ÷ 512. Get-Partition reports the Offset exactly in bytes (Fig 10).
2. The partitions on this GPT disk are laid out back-to-back, so Sector count = (next partition's start sector) − (this partition's start sector); End sector = next start − 1.
3. Bytes = sector count × 512, then ÷ 1024 three times to reach GB (binary, i.e. GiB).
4. Compare with Get-Partition, DiskPart and Get-Volume.

Why offsets are used. Get-Partition and DiskPart show partition sizes only as rounded values (for example "100 MB"), while the Offset column is exact. The exact size of each partition is therefore recovered from the difference between successive offsets, and then compared against the rounded values the OS prints. The offsets are all exact multiples of 512 and the disk size (512,105,932,800 B) divides exactly by 512 into 1,000,206,900 sectors, which supports the 512-byte sector size used.

6. Part D - Safe Partition Table Inspection

6.1 Windows method - PowerShell and DiskPart (Evidence 5)

```
C:\Windows\System32>diskpart

Microsoft DiskPart version 10.0.22621.1

Copyright (C) Microsoft Corporation.
On computer: DESKTOP-OSED05U

DISKPART>
```

Figure 8 - D1: diskpart - launching DiskPart (listing use only)

```
PS C:\Windows\system32> Get-Disk

Number F Serial Number                   HealthStatus         OperationalStatus
       r
       i
       e
       n
       d
       l
       y

       N
       a
       m
       e
------ - -------------                   ------------         -----------------
0      A [redacted]                      Healthy              Online


PS C:\Windows\system32>
```

Figure 9 - D1: PowerShell Get-Disk (serial masked) - Disk 0 Healthy, Online

```
PS C:\Windows\system32> Get-Partition


   DiskPath: \\?\scsi#disk&ven_acer&prod_ssd#4&28a914d7&0&000000#{53f56307-b6bf-11
d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                        Size Type
---------------  ----------- ------                        ---- ----
1                            1048576                     100 MB System
2                            105906176                    16 MB Reserved
3                C           122683392                233.11 GB Basic
4                D           250420920320             243.1 GB Basic
5                            511453429760                625 MB Recovery


PS C:\Windows\system32>
```

Figure 10 - D1: PowerShell Get-Partition - partition numbers, offsets, sizes, types [Evidence 5]

```
PS C:\Windows\system32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining
----------- ------------ -------------- --------- ------------ ----------------- -
D           Local Disk:  NTFS           Fixed     Healthy      OK                B
E                        Unknown        CD-ROM    Healthy      Unknown           B
C                        NTFS           Fixed     Healthy      OK                B
                         FAT32          Fixed     Healthy      OK                B
                         NTFS           Fixed     Healthy      OK                B


PS C:\Windows\system32>
```

Figure 11 - D1: PowerShell Get-Volume - volumes and file systems

```
DISKPART> list disk

  Disk ###  Status         Size     Free     Dyn  Gpt
  --------  -------------  -------  -------  ---  ---
  Disk 0    Online          476 GB  2048 KB        *

DISKPART>
```

*Figure 12 - D1: diskpart list disk - Disk 0, 476 GB, GPT column marked * [Evidence 5]*

```
DISKPART> select disk 0

Disk 0 is now the selected disk.

DISKPART>
```

Figure 13 - D1: diskpart select disk 0

```
DISKPART> list partition

  Partition ###  Type              Size     Offset
  -------------  ----------------  -------  -------
  Partition 1    System             100 MB  1024 KB
  Partition 2    Reserved            16 MB   101 MB
  Partition 3    Primary            233 GB   117 MB
  Partition 4    Primary            243 GB   233 GB
  Partition 5    Recovery           625 MB   476 GB

DISKPART>
```

Figure 14 - D1: diskpart list partition - Partitions 1-5 with size and offset [Evidence 5]

Partition	Type	Offset (Get-Partition, bytes)	Size shown	DiskPart offset	Drive letter / file system
1	System (EFI System Partition)	1,048,576	100 MB	1024 KB	none / FAT32
2	Reserved (MSR)	105,906,176	16 MB	101 MB	none / no file system
3	Basic (Primary)	122,683,392	233.11 GB	117 MB	C: / NTFS
4	Basic (Primary)	250,420,920,320	243.1 GB	233 GB	D: / NTFS
5	Recovery	511,453,429,760	625 MB	476 GB	none / NTFS

Interpretation. DiskPart marks Disk 0 with an asterisk in the Gpt column, so the disk uses GPT. The layout is the standard Windows UEFI layout: EFI System Partition, Microsoft Reserved partition, the C: Windows volume, a data volume, and a Recovery partition. DiskPart's offsets (1024 KB, 101 MB, 117 MB, 233 GB, 476 GB) agree with Get-Partition's byte offsets (1,048,576 ÷ 1024 = 1024 KB; 105,906,176 ÷ 1,048,576 = 101.0 MB; 122,683,392 ÷ 1,048,576 = 117.0 MB; 250,420,920,320 ÷ 1024^3 = 233.2 GB; 511,453,429,760 ÷ 1024^3 = 476.3 GB). The "Free 2048 KB" beside the disk in DiskPart is unallocated space on the disk; no partition was created or changed.

6.2 Linux/Kali method - read-only inspection

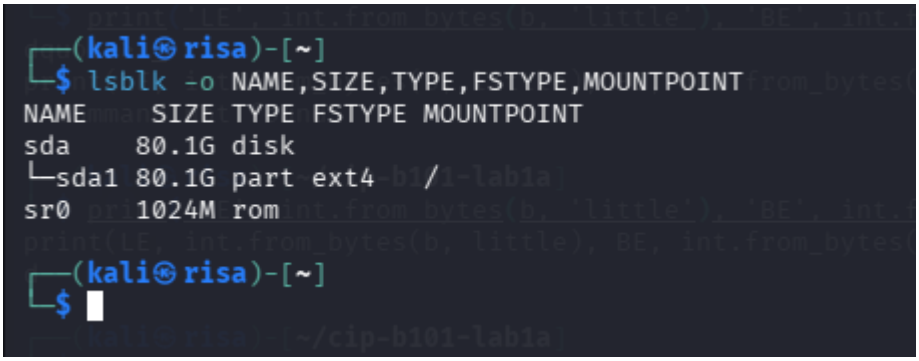


Figure 15 - D2: Kali lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT

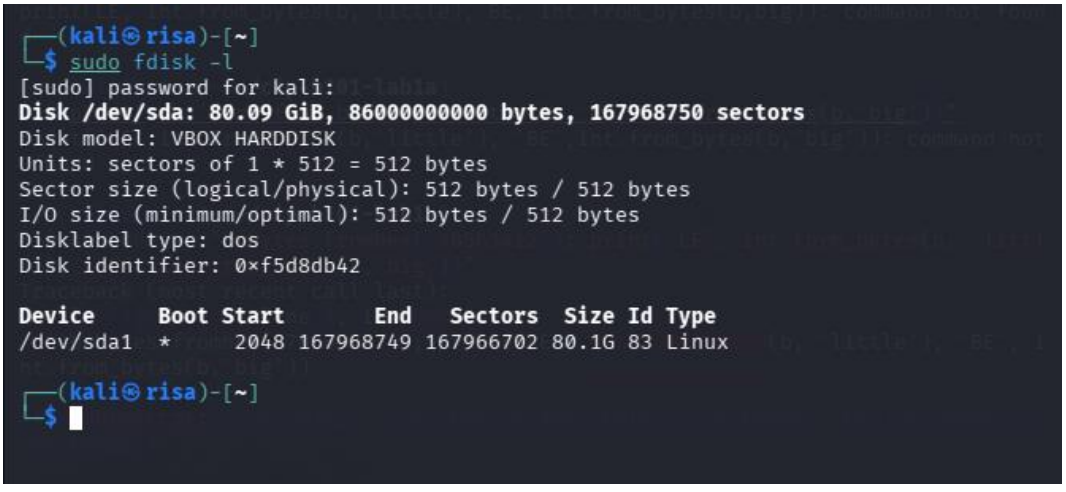


Figure 16 - D2: Kali sudo fdisk -l - exact Start, End and Sectors [Evidence 5]

```
┌──(kali㉿risa)-[~]
└─$ sudo parted -l
Model: ATA VBOX HARDDISK (scsi)
Disk /dev/sda: 86.0GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:

Number  Start   End     Size    Type     File system  Flags
 1      1049kB  86.0GB  86.0GB  primary  ext4         boot

┌──(kali㉿risa)-[~]
└─$
```

Figure 17 - D2: Kali sudo parted -l - partition table type msdos

```
┌──(kali㉿risa)-[~]
└─$ blkid

┌──(kali㉿risa)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            883M     0  883M   0% /dev
tmpfs           198M 1004K  197M   1% /run
/dev/sda1        79G   17G   58G  23% /
tmpfs           986M  4.0K  986M   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           986M  364K  986M   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           198M  108K  198M   1% /run/user/1000

┌──(kali㉿risa)-[~]
└─$
```

Figure 18 - D2: Kali blkid and df -h

Interpretation. The Kali VM has one virtual 80.09 GiB disk (VBOX HARDDISK, 86,000,000,000 bytes = 167,968,750 sectors) with disklabel type “dos”, i.e. an MBR partition table, and one Linux ext4 partition /dev/sda1 flagged bootable. The partition starts at sector 2048, leaving sectors 0–2047 (1 MiB) before it; sector 0 holds the MBR itself. blkid was run without sudo, so it printed nothing (it needs root to read device signatures); the file-system type was instead confirmed by lsblk -f and df -Th (Part F). No write operation was performed.

7. Part E - MBR, GPT and Boot Process

7.1 PC boot sequence

1. Power-on - the power supply stabilises and the CPU starts executing firmware code at a fixed address.
2. POST (Power-On Self-Test) - firmware tests CPU, RAM and basic devices; failures stop the boot or give beep codes.
3. BIOS/UEFI firmware - initializes hardware and reads its settings from CMOS/NVRAM (boot order, boot mode, Secure Boot).

4. Boot device selection - the first bootable device in the boot order is chosen.

5. Boot record - Legacy BIOS loads the MBR (sector 0) whose 446-byte boot code finds the active partition; UEFI reads the GPT, finds the EFI System Partition (FAT32) and runs an. efi boot application.

6. Boot loader / boot manager - Windows Boot Manager (bootmgfw.efi) or GRUB; offers any OS choice and loads the OS loader.

7. Kernel loading - the kernel and boot-start drivers are loaded into memory and take over the hardware.

8. Operating-system control - services start, the user logs on, and logs and user artifacts begin to accumulate.

Flow: Power-on → POST → BIOS/UEFI (CMOS/NVRAM settings) → MBR or GPT/EFI boot record → Boot loader/manager → Kernel → OS control

7.2 Boot mode of this system (Evidence 8)

See Figure 1 (BIOS Mode, Secure Boot State), Figure 3 (Boot Device) and Figure 12 (GPT marker).

Evidence	Observation
msinfo32 "BIOS Mode" (Fig 1)	UEFI
msinfo32 "Secure Boot State" (Fig 1)	Off
Boot device (Fig 1, 3)	\Device\HarddiskVolume1 - the first volume on the disk, i.e. the 100 MB EFI System Partition (Partition 1)
DiskPart list disk (Fig 12)	Gpt column marked * → Disk 0 is GPT
Partition 1 type (Fig 10, 14)	"System" (EFI System Partition), FAT32 (Fig 11)

Conclusion. The Windows host boots in UEFI mode from a GPT disk, which is the normal pairing: UEFI firmware reads the GPT and starts the Windows Boot Manager from the FAT32 EFI System Partition. Secure Boot is switched off, which is worth noting because it means the firmware does not enforce signature checks on boot loaders. The Kali VM, by contrast, uses an MBR ("dos") disk with a boot flag on /dev/sda1 (Fig 16, 17); an MBR disk is normally booted by legacy BIOS-style firmware, although the VM's firmware type was not captured in this lab.

7.3 MBR vs GPT (with observed examples)

Feature	MBR	GPT	Observed in this lab
Age / design	Early-1980s scheme for legacy BIOS	Modern scheme, part of the UEFI specification	Kali VM = MBR; Windows host = GPT
Where it lives	Sector 0: 446 B boot code + 4 × 16 B partition entries + 0x55AA signature	Protective MBR in LBA 0, primary header at LBA 1, entries from LBA 2, backup header and entries at disk end	Kali partition starts at sector 2048; 5 entries on Windows disk
Partitions	4 primary (or 3 + extended)	128 entries by default	Windows: 5 partitions
Size limit	≈ 2 TB (32-bit sector addresses × 512 B)	Very large (slide: ≈ 9.4 ZB)	Both disks far below either limit
Integrity	No checksum, no backup copy	CRC32 checksums and a backup table	–
Boot mode	Legacy BIOS	UEFI	Windows: UEFI + GPT
Forensic issues	Everything critical sits in sector 0; damage or a bootkit there changes how the disk is read; gaps between partitions can hide data	Protective MBR can mislead MBR-only tools; the backup structures help recovery and validation; GUIDs identify partition types	Partition type shown as “dos” vs “GPT *”

8. Part F - File System Identification and Interpretation

```
C:\Windows\System32>wmic logicaldisk get caption,filesystem,size,freespace,volumena
me
Caption  FileSystem  FreeSpace    Size          VolumeName
C:       NTFS        71302930432  250298232832
D:       NTFS        82787995648  261031456768  Local Disk:
E:


C:\Windows\System32>
```

Figure 19 - F: wmic logicaldisk get caption,filesystem,size,freespace,volumename [Evidence 9]

```
PS C:\Windows\system32> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining
----------- ------------ -------------- --------- ------------ ----------------- -------------
D           Local Disk:  NTFS           Fixed     Healthy      OK                          B
E                        Unknown        CD-ROM    Healthy      Unknown                     B
C                        NTFS           Fixed     Healthy      OK                          B
                         FAT32          Fixed     Healthy      OK                          B
                         NTFS           Fixed     Healthy      OK                          B


PS C:\Windows\system32>
```

Figure 20 - F: PowerShell Get-Volume - file system type of each volume [Evidence 9]

```
┌──(kali㉿risa)-[~]
└─$ lsblk -f
NAME FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNT
POINTS
sda
└─sda1
     ext4   1.0   root  894e89ec-b178-42b7-a450-6e969a2aa8fe   57.5G    21% /
sr0

┌──(kali㉿risa)-[~]
└─$
```

Figure 21 - F: Kali lsblk -f - FSTYPE, version, label, UUID [Evidence 9]

```
┌──(kali㉿risa)-[~]
└─$ df -Th
Filesystem     Type      Size  Used Avail Use% Mounted on
udev           devtmpfs  883M     0  883M   0% /dev
tmpfs          tmpfs     198M 1004K  197M   1% /run
/dev/sda1      ext4       79G   17G   58G  23% /
tmpfs          tmpfs     986M  4.0K  986M   1% /dev/shm
none           tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-journald.
service
tmpfs          tmpfs     986M  376K  986M   1% /tmp
none           tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-resolved.
service
none           tmpfs     1.0M     0  1.0M   0% /run/credentials/getty@tty1.servic
e
tmpfs          tmpfs     198M  108K  198M   1% /run/user/1000

┌──(kali㉿risa)-[~]
└─$
```

Figure 22 - F: Kali df -Th - file-system type and usage [Evidence 9]

Volume / partition	File system	Size	Role
C: (Partition 3)	NTFS	250,298,232,832 B (233.11 GiB)	Windows system volume
D: (Partition 4)	NTFS	261,031,456,768 B (243.10 GiB)	Data volume "Local Disk"
Partition 1 (no letter)	FAT32	100 MB	EFI System Partition (boot files)
Partition 5 (no letter)	NTFS	625 MB	Windows Recovery
E:	Unknown (no media)	–	CD-ROM drive
Kali /dev/sda1	ext4 (v1.0), label "root"	79G usable of 80.1G; 17G used (23%)	Linux root file system "/"

Identification notes. Get-Volume lists the volumes in this order: D (NTFS), E (CD-ROM, Unknown), C (NTFS), one unlettered FAT32, one unlettered NTFS. The unlettered FAT32 volume matches the 100 MB EFI System Partition (the only FAT32 partition expected on a UEFI Windows disk) and the unlettered NTFS volume matches the 625 MB Recovery partition; this pairing is an inference from sizes and order because Get-Volume did not print the size column legibly. The Microsoft Reserved partition has no file system and so appears in no volume list.

Why file systems matter in digital forensics. The file system defines how data is organised and tracked on a partition. It decides where file names, timestamps and permissions are stored, how deleted files are marked and whether their data can be recovered, whether slack space or alternate data streams exist, and which journals ($LogFile, $UsnJrnl on NTFS; the ext4 journal) can reconstruct recent activity. Forensic tools must recognise the file system to parse a volume, recover deleted items and build timelines, and the file system type guides tool choice and acquisition decisions. Here NTFS (MFT entries, MACB timestamps, journals) is the key source on the Windows volumes, FAT32 holds only boot files on the EFI partition, and ext4 (inode-based metadata, journal) applies to the Linux VM.

File system	Common use	Forensic importance
FAT/FAT32	Removable media, EFI System Partition	Simple structure; easy recovery of deleted entries; 4 GB file limit
NTFS	Modern Windows	MFT records, MACB timestamps, permissions, alternate data streams, journals
ext2/3/4	Linux	Inode-based metadata; journalling on ext3/4; deletion and inode-reuse affect recovery
XFS / JFS / btrfs	Linux / Unix-like	Journalling, snapshots and copy-on-write change recovery and metadata interpretation
HFS+	Older macOS	Key in Mac forensics; replaced by APFS

9. Part G – Final Practical: Partitions 1 and 2

Source of partition-table evidence: Get-Partition (Fig 10) and DiskPart list partition (Fig 14) on Disk 0. Partition style: GPT (Fig 12). Disk size 512,105,932,800 bytes = 1,000,206,900 sectors of 512 bytes. Figures 10 and 14 are the evidence for both partitions.

Task	Partition 1	Partition 2
Identity (Fig 10, 14)	System – EFI System Partition	Reserved – Microsoft Reserved (MSR)
Offset (bytes)	1,048,576	105,906,176
Start sector (offset ÷ 512)	2,048	206,848
Next partition starts at sector	206,848	239,616 (122,683,392 ÷ 512)
End sector (next start − 1)	206,847	239,615
Sector count	204,800	32,768
Bytes per sector	512	512
File system	FAT32	None

9.1 Step-by-step calculation

Partition 1 - EFI System Partition

Step	Working	Result
1. Start sector	1,048,576 ÷ 512	2,048
2. Next start	105,906,176 ÷ 512	206,848
3. Sector count	206,848 − 2,048	204,800 sectors (end sector 206,847)
4. Bytes	204,800 × 512	104,857,600 bytes
5. KB	104,857,600 ÷ 1024	102,400 KB
6. MB	102,400 ÷ 1024	100 MB
7. GB	100 ÷ 1024	0.0977 GB (0.09765625)

Partition 2 - Microsoft Reserved

Step	Working	Result
1. Start sector	105,906,176 ÷ 512	206,848
2. Next start	122,683,392 ÷ 512	239,616
3. Sector count	239,616 − 206,848	32,768 sectors (end sector 239,615)
4. Bytes	32,768 × 512	16,777,216 bytes
5. KB	16,777,216 ÷ 1024	16,384 KB
6. MB	16,384 ÷ 1024	16 MB
7. GB	16 ÷ 1024	0.0156 GB (0.015625)

9.2 Verification against operating-system information

	Calculated	Get-Partition (Fig 10)	DiskPart (Fig 14)	Get-Volume (Fig 11)	Result
Partition 1	100 MB (0.0977 GB; 104,857,600 B)	100 MB	100 MB	FAT32 volume present	Match
Partition 2	16 MB (0.0156 GB; 16,777,216 B)	16 MB	16 MB	no volume (no file system)	Match

Supplementary check on the larger partitions (same method, difference of successive offsets):

Partition	Calculation	Calculated size	OS-reported	Difference
3 (C:)	(250,420,920,320 − 122,683,392) = 250,298,236,928 B ÷ 1024^3	233.108 GB	Get-Partition 233.11 GB; wmic C: 250,298,232,832 B (233.108 GB)	4,096 B (NTFS volume is slightly smaller than its partition)
4 (D:)	(511,453,429,760 − 250,420,920,320) = 261,032,509,440 B ÷ 1024^3	243.105 GB	Get-Partition 243.1 GB; wmic D: 261,031,456,768 B (243.104 GB)	1,052,672 B (≈ 1 MiB): NTFS overhead and any alignment gap before Partition 5, which this method counts into Partition 4

9.3 Explanation of differences

- No difference for Partitions 1 and 2: both figures are exact powers of two (100 MiB and 16 MiB), so the calculated and reported values agree exactly. Windows labels binary megabytes as “MB”, which is why a hand calculation using 1024 gives exactly 100 and 16.
- GB vs GiB: dividing by 1024^3 gives binary gigabytes (GiB). In decimal units Partition 1 is 104.86 MB and Partition 2 is 16.78 MB, and the whole disk is 512.11 GB (the “512GB” in the model name) against 476.94 GiB (476 GB in DiskPart, which truncates).
- Rounding: Get-Partition and DiskPart round to whole MB/GB, so the calculated value is the more exact one.
- File-system overhead: a formatted volume (wmic/Get-Volume) reports slightly less than its raw partition because of file-system metadata and reserved sectors, as seen for C: (4,096 B) and D:.
- Partition 2 (MSR) has no file system and no drive letter, so only partition tools, not Get-Volume or wmic, show it.

Linux cross-check (MBR disk). /dev/sda1 on the Kali VM: start 2,048, end 167,968,749 (Fig 16). Sector count = 167,968,749 − 2,048 + 1 = 167,966,702 (matches fdisk). Bytes = 167,966,702 × 512 = 85,998,951,424; ÷ 1024^3 = 80.09 GiB (fdisk 80.1G, lsblk 80.1G); decimal = 86.0 GB (parted 86.0GB).

df -h shows 79G because ext4 reserves space for its own structures. This confirms the method on a second partition table type.

10. Conclusion and Summary (Evidence 10)

The preliminary examination profiled an HP Laptop 15-bs0xx running Windows 11 Pro (build 22621, x64) with an Intel Core i5-7200U, about 16 GB of RAM, and a single 512 GB Acer SSD (512,105,932,800 bytes, 1,000,206,900 sectors). The system boots in UEFI mode with Secure Boot off and the disk uses GPT, with five partitions: EFI System (FAT32), MSR, C: (NTFS), D: (NTFS) and Recovery (NTFS). A Kali Linux VM used an MBR disk with one ext4 partition, providing a direct contrast between MBR and GPT.

Using Get-Partition offsets and DiskPart output, Partition 1 was calculated as sectors 2,048–206,847 (204,800 sectors = 104,857,600 bytes = 100 MB = 0.0977 GB) and Partition 2 as sectors 206,848–239,615 (32,768 sectors = 16,777,216 bytes = 16 MB = 0.0156 GB). Both match the sizes reported by the operating system exactly; the small differences seen on the larger volumes are explained by GiB/GB units, rounding and file-system overhead.

This matters in digital forensics because evidence lives in several layers that corroborate each other; volatile process data must be captured before shutdown; the partition table and boot mode (MBR/GPT, BIOS/UEFI) determine how a disk is interpreted and where hidden or unallocated data may sit; and the file system determines which metadata, timestamps and journals are available for recovery and timelines. Every step was performed read-only, so the system under examination was not altered by the analysis, although a running system still changes by itself (compare the free-space values in Figures 6 and 19).

Appendix A - Required evidence checklist

No.	Required evidence	Figure(s)	Included
1	System Information / systeminfo	1, 2, 3, 4	Yes
2	Disk model and capacity	5	Yes
3	Logical volume / file system information	6, 11	Yes
4	Running processes and PIDs	7	Yes
5	Disk and partition table inspection	9–14 (Windows), 15–18 (Linux)	Yes
6	Partition 1 calculation and evidence	10, 14; §9.1	Yes
7	Partition 2 calculation and evidence	10, 14; §9.1	Yes
8	BIOS/UEFI boot mode	1, 3, 12; §7.2	Yes
9	File system identification	19, 20, 21, 22	Yes
10	Final summary and conclusion	Section 10	Yes